DNS Walkthrough:

Domain Name System (DNS) is a system that helps improve user's ease when visiting a website. Instead of having to remember the IP address (e.g. 192.x.x.x) to visit a website, a user can simply use the domain name (e.g. google.com). DNS enables this by mapping the name to the IP address assigned to that domain.

Currently, my website is hosted on Vercel and is accessible through a free *.vercel.app* URL. If I later connect a custom domain to my Vercel project, DNS will allow that domain to point to my webpage.

When a user enters my website's link, the following steps take place to display the webpage:

1. The user enters my website's domain into the browser.
2. The user's browser checks for a cached DNS result. If not, the browser sends a request to a DNS recursor.
3. The DNS recursor sends a query to DNS root nameserver to find the nameservers responsible for the domain's TLD (e.g. .com).
4. The recursor then queries the relevant TLD nameserver, which directs it towards the authoritative nameserver responsible for the target domain.
5. The recursor requests the authoritative nameserver which holds the DNS records for the domain.
6. The appropriate DNS record is returned to the recursor which returns this result to the user's browser.
7. The user's browser then uses this information to connect to the relevant hosting service. In my case it would be Vercel.
8. Vercel then serves the website to the user over HTTPS.

A CNAME (Canonical Name) record is used to point one hostname to another. Instead of directly storing the IP address, it tells DNS to look up another hostname.

For example, if I use a custom subdomain name, I will use a CNAME record to point that subdomain towards the hostname provided by Vercel. The exact CNAME value is provided by Vercel when the custom domain is configured.